



Two perceptions of software maintenance performed by an on-site contractor

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ABSTRACT

Software maintenance is a difficult task under the best of circumstances. Having work performed by an on-site contractor adds an additional layer of complexity to the customer's task. This type of relationship places greater emphasis on formal work procedures and detailed reports of the work in progress. It also promotes the use of performance norms for evaluating contractor performance. These factors are all on the positive side. However, such a relationship also calls for a special awareness of contractor ploys calculated to increase their performance evaluation.

From the contractor's point of view, being on-site imposes a more disciplined environment and places special importance on the manner and means of dealing with the customer. Another special feature is that the contractor receives formal feedback from the users, through periodic performance evaluations, indicating how well the software maintenance group measures up to expectations.

This paper describes the lessons learned by one customer and one on-site contractor.

CUSTOMER'S VIEW

contracting is likely to become more pervasive due to basis in the federal government¹ (à la the Office of Management and Budget Circular A-76) as well as in state agencies and municipalities. An account is given here of a large data processing facility with extensive experience acting out software maintenance has learned to cope. The data processing installation is a large, multisystem government facility, comprised of a mix of manufacturers and included are on-line systems, database systems, batch and intercomputer systems. Types of hardware include IBM (370, 4341), UNIVAC (1100, Varian), SEL (32), Honeywell (Sigma). Altogether, there are 20 stand-alone systems of the applications run on these systems deal with financial data; however the operational mode is akin to a multistep production process. Other applications relate to production control, cost accounting, inventory control, and equipment maintenance.

Centralized vs. Decentralized Support

Initially, our technical control over the work performed by the contractor was split along application and functional lines. It was split into several independently run units, both on our side and on the contractor's side. However, we exercised overall financial stewardship over the contractor's activities for evaluation of performance.

Our current arrangement, while providing us with a close working relationship and a strong grasp of the technical details, naturally led to parochial viewpoints on both sides. If a key system was in one area resigned, contractor personnel could easily call upon another area for temporary assistance because of reluctance by the latter to dilute their level of support. Support problems, resulting from poor management, understaffed or inadequate numbers of personnel, and the tendency to be prolonged and not pursued aggressively. In early 1982, we reorganized to centralize all software maintenance within a single unit. The contractor's organization was reconstituted on a centralized basis. A number of fits—some obvious and some not so obvious—were identified, including

uniform reporting of maintenance activities
uniform and tighter configuration control
more effective communications channels
improved response in correcting or resolving problems
separation and independence of programming and testing groups

6. improved documentation (due to configuration control oversight)
7. more effective control within the program library
8. more effective establishment of priorities and better allocation of resources
9. more consistent manner evaluation of contractor performance and determination of award fees
10. more availability of the information to build a centralized database for deriving work performance metrics

Establishing an Effective Working Relationship

Because software maintenance cannot easily or readily be translated into a set of well-defined products, the connection between customer and contractor needs special emphasis. This is a critical factor in determining the quality and cost effectiveness of the support provided by the contractor. The key elements characterizing the customer-contractor relationship can be labeled as the three Cs: credibility, coverage, and clout. Credibility hinges largely on the competence of customer personnel. The level of competence should be such as to convince the contractor personnel that the customer is fully aware of work factors—do's and don'ts—and that expectations of the contractor's performance are reasonable. Customers need to be candid in their dealings with contractors and to view them as co-workers rather than as subordinates or in a potentially adversarial position. Unfortunately, this can easily lead to a "chummy" relationship, which can be harmful. Customer personnel should not forget that, basically, this is a business relationship that calls for critical assessment of the contractor's performance. In particular, customer personnel need to distinguish between legitimate extenuating circumstances and groundless excuses. Otherwise, the contractor will not make a concerted effort to correct deficiencies.

Coverage, as used here, refers to the organizational or functional level at which the customer-contractor connection takes place, as well as the depth of reporting detail. Good starting points are the systems and methods for selecting and controlling the jobs to be performed. These come under the general heading of configuration control. In our case, a formal system was agreed upon that would govern the submission, review, disposition, and reporting of change requests. These include system and application software errors, deficiencies, enhancements, and new system releases. Also included are special tasks that compete for the programming group's resources.

At least weekly status meetings should be held and should include contractor line managers (the heads of system software, applications software, and testing), along with the customer technical monitors. Coverage is really a corollary of

credibility in that it is meant to ensure that the technical discussions are substantive and are more likely to flush out causes, rather than treat symptoms.

Clout is a two-edged sword. It can be and should be used both to reward the contractor for better-than-expected performance and to penalize for below-expected performance. One way to accomplish this is by way of a cost-plus-award-fee contract, with the award ranging from 0% to 10% of the cost. Expected performance results in a fee in the 4% to 6% range, thereby leaving ample allowance for award level variations based either on positive or negative factors. Another important consideration is the level of management—both sides—that is involved in or is made aware of the fee determination. On the customer side, this should mean the top person in charge of the data processing facility; on the contractor's side, it should mean at least one level above the on-site manager, depending on whether software maintenance represents part or all of the contract. If the latter, the involvement should be at least two levels above the contract manager.

What You Need to Know

What do customers need to know about the contractor's activities in order to monitor and evaluate the contractor's performance effectively? In our case, we have stipulated that reporting should be at the functional (or third) level with system and project reporting being the higher levels.

For our purposes functional reporting was broken down as follows: validation and assessment of the effects of a proposed change (prior to approval by the configuration control board); programming (analysis, coding, and unit testing); acceptance testing; and implementation. This breakdown is predicated on the objective of closely monitoring the work in progress so as to be conversant with current problems and to assess effectively whether proper and timely actions are being taken to resolve them.

Some might argue that on a routine basis it is only necessary to monitor the contractor's activities at either the system or project levels and thereby reduce the cost of monitoring. It is further argued that either periodic or unannounced audits can be made to determine the contractor's performance at the

functional level. The problem with this argument is, that substantive deficiencies are uncovered by an audit. Customer monitors are not in a position to assess independently whether the contractor is taking the proper corrective measures—and doing so in a timely manner. Waiting until the next audit takes place to make such a determination is not an effective way to deal with such problems.

It should not be inferred that effective monitoring requires an item-by-item review. One suggestion is to have an item placed in the margin of a report to highlight those items in which actual hours exceeded estimated hours or for which a data change was made since the last report period. This draws the monitor's attention to the items that require close observation and that should be accompanied by a written explanation. A complementary tactic is to specify the most important items in a separate report, which is distributed at a higher level of management than is the full detailed report.

Specifying the detailed items to be reported on is critical to the battle. The reports must be reviewed carefully for accuracy, completeness, and currency. Contractors are prone to adopt a casual attitude toward reporting if they are not held closely and consistently accountable for their report contents. Figures 1 and 2 are sample formats of monthly reports by system, showing, respectively, the current status of all work in progress and the actual hours expended by type of job.

Games Contractors Play

Wherever there are performance-type contracts, there is an inclination to "shade" the reporting of activities and expenses in a way that is advantageous to the contractor. Although this can, in reality, be a euphemism for fraudulent practices, it is more likely to manifest itself in more subtle and odious forms. Also, on-site contractors are less likely to engage in these practices than off-site contractors, because of the more personal relationship in the former case.

Noted here are both known and suspected tactics that contractors have used. These tactics represent an overgeneralization drawn from a number of different contractors.

NUMBER OF JOBS	OPERATING SYSTEMS			APPLICATIONS PROGRAMS			SPECIAL JOBS
	<u>FIXES</u>	<u>ENHANCE.</u>	<u>OTHER</u>	<u>FIXES</u>	<u>ENHANCE.</u>	<u>OTHER</u>	
Open - Beginning of Month							
Added During Month							
Closed During Month							
Open - End of Month							

Figure 1—Monthly change in work status

<u>TOTAL HOURS EXPENDED FOR</u>	<u>OPERATING SYSTEMS</u>			<u>APPLICATIONS PROGRAMS</u>			<u>SPECIAL JOBS</u>
	<u>FIXES</u>	<u>ENHANCE.</u>	<u>OTHER</u>	<u>FIXES</u>	<u>ENHANCE.</u>	<u>OTHER</u>	
Analysis							
Code and Unit Test							
Implementation							
 <u>FOR CLOSED JOBS</u>							
Analysis Hours -	Est.						
	Act.						
Code & UT Hours -	Est.						
	Act.						
Implementation Hours- Est.							
	Act.						
Total Hours -	Est.						
	Act.						

Figure 2—Manhours expended during month

Creative bookkeeping

To prevent actual hours from exceeding estimated hours on a given job, time is charged to "miscellaneous." A variation of creative bookkeeping is where the contractor in the process of trying to correct an error takes a shortcut, e.g., bypassing testing, in order to stay on schedule. Should this in turn cause additional errors, these are reported as new errors and are disposed of expeditiously. This, of course, leads to "favorable" measured performance.

Technical obfuscation

When analysis or diagnosis of a persistent problem does not turn up anything definite, or when an embarrassing event occurs, the contractor might try to talk his way around it. Jargon and vague but technically imposing reasons might be offered to convince customer monitors that the problem is not due to any fault of the contractor.

All in the family

Here, contractors try to be particularly responsive to the customer monitor's pet projects. This is coupled with ego-

boosting tactics, which together are an attempt to foster the impression that "we are all family" and we ought to be protective of the other party's interests. A variation of this game is to seek the company monitor's advice and suggestions about how to handle a given problem. This tends to compromise the company monitor's objectivity in assessing the contractor's performance.

End-around play

Should the customer monitors prove rather astute in dealing with the contractor's games, or if the customer monitors are frequently critical of the contractor's performance, a play can be made to a higher level of management. An attempt is made first to establish a close liaison with higher management and then to convince them that the monitors are biased and unreasonable.

Old standbys

Briefly noted here are the more familiar excuses and tactics used by software personnel.

1. overly generous padding of estimates to perform jobs

2. blame it on the vendor's documentation
3. blame it on the operating system
4. blame it on the hardware
5. blame it on the person no longer employed by the contractor

Performance Measurement

Under an incentive-type contract, it is necessary to face the issue of performance measurement squarely. First to be addressed is the formulation of which elements and factors are to be evaluated and measured. The candidate elements are those from which one can derive the desired factors. Examples of such factors include management, productivity, responsiveness, timeliness, communication, planning, and initiative. Factors such as management, communication, and planning are highly subjective in nature and are evaluated in an indirect or on an event basis. Others, such as productivity, responsiveness, and timeliness, are adaptable to objective measurement, and these are the ones discussed herein.

Before qualifying a set of metrics for performance evaluation, it is necessary to define and establish a database. In our case, pertinent information is collected from all jobs including application software changes, operating systems maintenance, and special software tasks. Information about these jobs is collected from the individual programmers, and entered into a database. Weekly reports compiled from this information are carefully reviewed both by the contractor supervisors and the monitors to assure complete reporting and overall accountability. A list of the metrics that we observe is shown in Table I.

Each of the metrics in the table can be further categorized by computer system, language, type (i.e., systems or application software), and so forth. Such breakdowns enable comparisons to be made within the given category; e.g., how does the average time per fix for system A compare with that for system B?

After a sufficient amount of time has elapsed to compile a substantial database and to analyze and interpret the derived metrics, the final step can begin. This is to establish the norms for each of the selected metrics. Here again, contractor personnel should participate in this determination in order to arrive at a set of norms that is deemed to be fair and reasonable to both parties.

Such objective performance measures can be weighed and coupled with the subjective factors referred to earlier so as to arrive at the contractor's overall technical performance assessment.

THE CONTRACTOR'S VIEW

Interfacing with Customer Personnel

The role of the software professional within a company that performs facility management services is somewhat different from that of a programmer nestled comfortably in a corporate structured arena. Being on-site readily exposes a casual or

TABLE I—Performance Metrics

Metric	Derivation
Average time to make a fix or enhancement	Total hours for analysis, coding, unit testing divided by total number of fixes and enhancements
Average elapsed time to make a fix or enhancement	No. of days from start to implementation divided by total number of fixes and enhancements
Actual vs. estimated time per fix or enhancement	Total actual hours divided by total estimated hours for fixes and enhancements
Elapsed time of highest priority fixes vs. others	Average elapsed time to make highest priority fixes divided by average of elapsed time to make all other fixes
Trend analysis of reported software failures	Comparison of distributions of failure occurrences for different systems
Correlation of number of fixes with size of program	Dependent variable is the number of fixes for each program; independent variable is the size of each program
Standard deviation of estimates of large vs. small programs	Standard error of actual vs. estimated hours for each fix, grouped by program size
Ratio of analysis time per fix to coding and unit testing	Total hours for analysis of all fixes divided by total hours for coding and unit testing

sloppily managed working group and calls for an awareness or presence that should be calculated to command the respect of the customer. Sloppy personal demeanor, unoccupied desks, persons reading newspapers, and so on, are perceived by the customer as indicators that the contractor is unreliable, unprofessional, or underworked. In effect, the contractor has two "bosses"—the on-site customer as well as company management. This presents a unique dilemma—how to please both factions and maintain proper professional perspective (and sanity) in successfully fulfilling job requirements.

Acquiring the confidence of customer-monitoring personnel is an important goal that must be achieved quickly if successful performance ratings are to be attained. The ability to grasp the technical jargon and the complexities of the customer's subject matter makes customer communication a natural extension of the monitor's working environment.

When special requirements are addressed, the contractor should obtain customer concurrence on how the workload should be adjusted to satisfy all affected users. Too often, additional task requirements are accepted by the contractor without informing the customer of current manpower con-

straints and the effects of new tasks on current completion schedules. The contractor must not be afraid to oppose additional customer requests and should be prepared to convey to customer management that in reality there is no free lunch. When possible, suitable alternatives should be recommended.

Effective communication of task performance appraisals is an area that requires special contractor attention. The customer needs appropriate status information to provide a sufficient base for pointing out shortcomings, giving plaudits for tasks well done, and recommending an appropriate award fee. Formats for contractually required reports should be determined mutually, at the beginning of the contract, and should be reviewed periodically for possible alteration to respond to changing customer management reporting requirements. In addition to these reports, regularly scheduled status meetings between software management and key customer-management-technical-monitoring personnel should be established. These meetings, which are by design less formal and in the nature of committee sessions, are multipurpose. They not only provide a forum for presenting firsthand status information, but also are an excellent opportunity for discussing customer priorities and perceived deficiencies prior to their being written into the customer's evaluation report. Another helpful measure is to provide a self evaluation—representing the software management's view of task performance—to the customer for consideration in determining periodic award fees.

How Work is Divided and Allocated

As noted in the first part of this paper, we are a centralized software organization, responsible for maintaining more than 50 software systems functioning on more than 20 mainframes, and for all developmental work. Major functions are separated into applications programming, systems programming, and software acceptance testing. By definition, applications programmers are responsible for maintaining the production software (primarily FORTRAN coding, with some assembly language) and the systems programmers are the caretakers of all operating system software. Systems analysts, however, provide the necessary expertise for assuring the validity of both new and modified software through the development and execution of detailed acceptance test plans.

Because of the size of this organization—approximately 85 software professionals—the numerous specially developed computer systems, and the frequency of software changes attributable to data-related and user requirement variances, it is difficult to impose conventional software management techniques. An internal task-tracking system has been developed to monitor several hundred tasks ranging from discrepancy reports (something doesn't look right) to change requests (modifications to accommodate specific problems or requirements). Included within this range are customer-initiated tasks (often new requirements) and tasks generated internally by software management (usually related to normal maintenance activities, such as evaluating release tapes for existing operating systems). Due to the high volume of tasks, complexities of interorganizational interface, and management

requirements for up-to-date status reporting, a full-time administrator is employed to maintain and coordinate all transactions and report generation attributed to this tracking system.

Assignment of programmers to support each system can often be a difficult process. Software management must be prepared to evaluate the overall complexity of the system, be familiar with the intricacies of various program components, and be knowledgeable about the stability or volatility of the software. These variables are then matched against individual programmer experience profiles to determine the most appropriate manpower allocation.

Acquiring and Retaining Technical Personnel

Our typical maintenance programmer has almost five years of college training and more than six years of technical experience. Turnover, however, is surprisingly low in our case, because of an unusual phenomenon known as incumbency. Many of our software professionals have selected this area because of the nature of the work—it is highly scientific and very interesting; the physical plant is conveniently located and easily accessible; there is no charge for parking; etc. Even though the contract is bound by a prenegotiated amount of time, the technically oriented employee has little fear of losing a position due to contract expiration. Obviously, even under a new contractor, the job must continue to be performed. Who else, other than those currently doing the job, could satisfy customer requirements with no untoward effect on daily operations? Of course, if there is a new contract awarded, management must be sensitive to the apprehension programmers are likely to exhibit during the recompetition and, if necessary, the changeover periods.

Programmers, like many other skilled professionals, consider themselves creative and take special pride in developing "eternal" systems. There exists, then, an innate stigma attached to the label of "maintenance programmer." This is a difficult but not insurmountable hurdle for software management to overcome. One of the ways to maintain good personnel morale is by offering diversification in mainframes, operating systems, and programming languages. For example, in our case the opportunity to use FORTRAN, assembly language, or PL/I may be found on IBM (370/145 and 4341) using VS1, VM, or MVS; IBM (Series-1) using EDX; UNIVAC (1100/82) using EXEC-8 38R2; SEL (32/77 and 32/75) using RTM and MPX-32; VARIAN 77 using VORTEX; SIGMA 5/9 using BPM and CP-V; and various other special purpose image-processing systems.

Although the term maintenance is used to describe the main functions, many tasks require such extensive systems analysis prior to making appropriate changes that the programmer receives as much challenge and satisfaction as if the program was actually being developed. Another factor is training. In order to keep the staff abreast with state-of-the-art developments, management encourages formal vendor-supplied training classes. Specific analytical and systems-oriented techniques and skills are addressed in these courses. Attendance at user and general conferences is also an added

incentive provided to the programming professional for acquiring and dispensing information.

Dealing with Newly Developed Software

Almost all software maintenance groups encounter the problem of assuming responsibility for new software developed by another organization. In our case, this problem is compounded by the fact that the new programs are developed by another contractor. To deal effectively with this situation requires getting involved well before the software is delivered. Plans and interface definitions should be mutually agreed upon and include acceptance testing, documentation, and formal sessions for acquainting the maintenance personnel with the inner workings of each program.

The development of the acceptance test plan requires extensive communication between the maintenance and the development groups. Program design walk-throughs are highly recommended for this purpose, as well as for familiarizing the maintenance personnel with the software. This should be done prior to the turn-over of the program since afterwards development personnel are reassigned to other tasks and often are not easily accessible.

As on-site contractors, we need to be particularly concerned with the way information concerning our dealings with development personnel is presented to the customer. Group interaction problems, such as competing for computer time, should if possible be transparent to the customer. When these problems need to be brought to the customer's attention, it is best to avoid a finger-pointing session. Such sensitivity and awareness contribute measurably to harmonious relations with the customer.

CONCLUSION

Overall, the use of on-site contractors can be a viable and effective means for accomplishing software maintenance in a large data processing facility. To achieve these ends, however, calls for a proper appreciation by both the customer monitors and the contractor management personnel of the factors and considerations described herein.

REFERENCE

1. Office of Management and Budget Circular A-76. This circular has been incorporated into the Federal Acquisition Regulation as Subpart 7.3, effective April 1, 1984.